A necessary condition for a coprime kernel

Private continuation, phase 277. This obstruction concerns independent marked groups in arbitrary rank. The displayed example is expressly not a polytope monodromy group, and is not a counterexample to the finite-cover conjecture for polytopes.

Let $\pi : H \to G$ be an epimorphism of finite marked string groups, and suppose H has the intersection property. Take disjoint color sets I, J such that every mark in one set commutes with every mark in the other; it is enough that every cross distance exceeds one. Put $A = H_I$, $B = H_J$, and $K = \ker \pi$.

Proposition 1. The order of $G_I \cap G_J$ divides $|K|$. Consequently a C-cover with kernel order coprime to $|G|$ is possible only if all such separated parabolic intersections in G are trivial.

Proof. The subgroups A and B commute and intersect trivially, so multiplication identifies AB with $A \times B$. Let $D = \pi(A) \cap \pi(B)$. Define a homomorphism

$$f : K \cap AB \longrightarrow D, \qquad f(ab) = \pi(a).$$

The factors a, b are unique. If $ab \in K$, then $\pi(a) = \pi(b)^{-1} \in D$, so the map is well-defined. Commutation of A and B proves multiplicativity. For every $d \in D$, choose $a \in A$ and $b \in B$ with $\pi(a) = d = \pi(b)$. Then $ab^{-1} \in K \cap AB$ and $f(ab^{-1}) = d$, proving surjectivity. The first isomorphism theorem and Lagrange's theorem now show $|D| \mid |K \cap AB| \mid |K|$. Since $|D|$ also divides $|G|$, the coprime assertion follows. ■

This condition cannot be omitted from a higher-rank extension of the rank-four p-kernel theorem. Let $D_8 = \langle a, b \mid a^2 = b^2 = (ab)^4 = 1 \rangle$, with central involution $z = (ab)^2$. Use two copies, with marks (a_0, a_1) and (b_3, b_4), and define

$$G = (D_8 \times D_8)/\langle (z, z) \rangle \times C_2.$$

Mark the five factors in the order a_0, a_1, t, b_3, b_4, where t generates the final direct factor. Then $|G| = 64$. The five marks are independent: the abelianization of the central product is C_2^4, since the identified central subgroup already lies in the commutator subgroup, and the extra factor contributes one more independent coordinate. All required string commutations hold.

Both maximal contiguous end groups have order 32 and are isomorphic, with the indicated separated components, to $D_8 \times C_2 \times C_2$. They are C-groups; so are all their standard parabolics. But

$$G_{\{0,1\}} \cap G_{\{3,4\}} = \langle \bar{z} \rangle \cong C_2.$$

Proposition 1 implies that **every finite C-cover of this marked group has even kernel order**. In particular no odd-prime p-group kernel can work, regardless of its dimension or number of stages. The product $D_8 \times D_8 \times C_2$ itself is a finite C-cover with kernel of order2, so the example does not obstruct finite existence.

Why this is not a polytope counterexample is equally precise. In the monodromy action of an actual polytope, a word using only colors in I fixes every face of ranks outside I, at every flag. If an element also belongs to the subgroup on a disjoint set J, it fixes faces of every rank, and therefore fixes every flag. The monodromy permutation is the identity. Thus actual monodromy satisfies

$$G_I \cap G_J = 1 \quad \text{whenever } I \cap J = \emptyset,$$

with no need for the separated-color assumption. The example above violates this necessary condition.

The remaining finite-polytope problem must exploit this genuine flag-action condition when attempting a uniform p-kernel construction in higher rank. Extreme independence alone, sufficient in rank four, no longer suffices. No assertion of historical novelty for Proposition 1 or this central-product example is made without a separate literature check.

Exact finite model and a qualification about covers

The independent reader can reconstruct the example without identifying a small-group catalogue entry. Write an element of the order-64 group as $(v_0, v_1, v_3, v_4, z, t) \in \mathbb{F}_2^6$, with multiplication

$$(v, z, t)(w, d, u) = (v + w,\ z + d + v_1 w_0 + v_4 w_3,\ t + u).$$

The five marks are the coordinate vectors in positions v_0, v_1, t, v_3, v_4. The order-128 cover uses two central coordinates, with cocycles $v_1 w_0$ and $v_4 w_3$ separately; its quotient map adds those coordinates. The program `phase277_obstruction_check.py` checks both multiplication tables, all associativity triples, the marked epimorphism and its two-element kernel, all 1,024 parabolic intersections of the cover, and all 256 pairs in each end of the base group. This is a certificate for the displayed finite model. The divisibility assertion about every finite C-cover is Proposition 1's written proof, not a conclusion of those finite checks.

There is no claim that the order-128 cover is the only minimal cover. In particular, centrality of g_2 does not force centrality of its lift: the adjacent commutations with g_1 and g_3 need not survive in a cover. It would be incorrect to split every cover into three direct factors on the basis of the direct-product presentation of this particular base.